!**List Services**: LicensePlateRecognition, WeaponDetection, FireSmokeDetection, FloodDetection,
SuspiciousBehaviorDetection, GraffitiDetection, AlertManagement, VehicleAuthorization,
BodyPoseDetection, StreamManagement
!**Standard Mode**: **off**
!**List Modes**: **NORMAL**::Normal::**default**,
EMERGENCY::FireEmergency, **EMERGENCY**::ArmedThreatEmergency, **EMERGENCY**::FloodEmergency,
DEGRADED::ReducedCoverage, **DEGRADED**::SingleOperator
!**List Exceptions**: {**HARDWARE_EXCEPTION**::CameraOffline},
{**NETWORK_EXCEPTION**::StreamInterrupted}, {**NETWORK_EXCEPTION**::DashboardUnreachable},
{**NETWORK_EXCEPTION**::RadioCommunicationDown}, {**NETWORK_EXCEPTION**::MessageServiceDown},
{**NETWORK_EXCEPTION**::MessageBrokerDown},
{**SOFTWARE_EXCEPTION**::DetectionModelFailure}, {**SOFTWARE_EXCEPTION**::DatabaseUnavailable},
{**SOFTWARE_EXCEPTION**::OCRFailure}, {**SOFTWARE_EXCEPTION**::PoseModelFailure},
{**ENVIRONMENT_EXCEPTION**::FireDetected}, {**ENVIRONMENT_EXCEPTION**::FloodDetected},
{**ENVIRONMENT_EXCEPTION**::WeaponDetected}, {**ENVIRONMENT_EXCEPTION**::SuspiciousBehaviorDetected},
{**ENVIRONMENT_EXCEPTION**::GraffitiDetected}

use case: MonitorCampusSurveillance
scope: SurveillanceSystem
level: **SUMMARY**
intention: "The Security Coordination Unit intends to monitor the university campus for security threats and incidents using the CSIS surveillance system."
multiplicity: "Multiple operators, security agents, and supervisors use the system simultaneously across shifts."
primary actor: **HUMAN**::Operator
main success scenario:

1. [AuthenticateUser] "Operator logs in to the System."
2. [MonitorCameraFeeds] "Operator monitors real-time camera feeds through the System."

3. [RespondToIncident] "Operator responds to incidents detected by the System."
use case ends in: **SUCCESS**

use case: ManageCampusSecurity
scope: SurveillanceSystem
level: **SUMMARY**
intention: "The supervisor intends to oversee and coordinate security operations across the university campus."
multiplicity: "Two supervisors manage the Security Coordination Unit."
primary actor: **HUMAN**::Supervisor
main success scenario:

1. [AuthenticateUser] "Supervisor logs in to the System."
2. "Supervisor reviews active alerts and event history on the System."
3. "Supervisor coordinates field agent deployments through the System."
4. "Supervisor generates reports and adjusts rule parameters in the System."

use case ends in: **SUCCESS**

use case: AuthenticateUser
scope: SurveillanceSystem
level: **USER_GOAL**
intention: "The operator or supervisor intends to authenticate to access the CSIS dashboard."
multiplicity: "Multiple users can authenticate simultaneously."
primary actor: **HUMAN**::Operator
main success scenario:

1. "System presents login screen to Operator."
2. "Operator enters credentials into System."
"System validates the credentials against the user database."
3. "System grants Operator access to the Dashboard."

use case ends in: **SUCCESS**

extensions:

alternative for 3:
"Operator enters invalid credentials."
3a. "System denies access and requests Operator to re-enter credentials."
use case continues at step: 2

use case: MonitorCameraFeeds
scope: SurveillanceSystem
level: **USER_GOAL**
intention: "The operator intends to monitor live camera feeds from across the campus for security threats."
multiplicity: "Three operators per shift monitor feeds from 100 or more cameras simultaneously."
primary actor: **HUMAN**::Operator
secondary actor: **SENSOR**::Camera::1..*
main success scenario:

1. [EstablishCameraConnections] "System establishes RTSP connections with Camera."
2. [ProcessVideoFrames] "System processes video frames from Camera through detection models."
3. [DisplayAlerts] "System displays detected events to Operator on the Dashboard."

"Operator reviews alerts and takes appropriate action."
use case ends in: **SUCCESS**

extensions:

exception for (1-3):
(1-3)a. {**ENVIRONMENT_EXCEPTION**::FireDetected} #Fire or smoke detected on campus. System initiates fire emergency protocol.#
use case ends in: **FAILURE**

exception for (1-3):
(1-3)b. {**ENVIRONMENT_EXCEPTION**::WeaponDetected} #Weapon detected on campus. System initiates armed threat protocol.#
use case ends in: **FAILURE**

exception for (1-3):
(1-3)c. {**ENVIRONMENT_EXCEPTION**::FloodDetected} #Flooding detected on campus. System initiates flood emergency protocol.#
use case ends in: **FAILURE**

use case: RespondToIncident
scope: SurveillanceSystem
level: **USER_GOAL**
intention: "The operator and security team intend to respond to a detected security incident."
multiplicity: "Multiple incidents can be responded to concurrently by different agents."
primary actor: **HUMAN**::Operator

secondary actor: **HUMAN**::SecurityAgent, **HUMAN**::Supervisor, **SOFTWARE**::HubOfEvents, **SOFTWARE**::SystemClock
main success scenario:
1. "Operator reviews alert details on the System including annotated video snapshot."
2. "Operator confirms the alert on the System and updates status to In Progress."
3. "System sends notification to SecurityAgent via radio and messaging application."
4. "SecurityAgent is dispatched to the incident location by System."
5. "SecurityAgent reports situation assessment back to System."
6. "Operator updates alert status to Closed on the System."

use case ends in: **SUCCESS**

extensions:

alternative for 2:

"Operator determines the alert is a false positive."

2a. "Operator dismisses the alert on the System and updates status to Dismissed."

use case ends in: **SUCCESS**

exception for 3:

3a. {**NETWORK_EXCEPTION**::RadioCommunicationDown} #Radio communication with SecurityAgent is unavailable.#

use case ends in: **FAILURE**

exception for 3:

3b. {**NETWORK_EXCEPTION**::MessageServiceDown} #Messaging application for SecurityAgent is unavailable.#

use case ends in: **FAILURE**

alternative for 5:

"Incident requires external emergency services."

5a. [EscalateToEmergencyServices] "Supervisor contacts emergency services through System."

use case continues at step: 6

context-awareness:

CA001: "The system selects response protocol by receiving incident type classification from HubOfEvents."

CA002: "The system prioritizes incident response by receiving current time of day from SystemClock."

use case: EstablishCameraConnections
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to establish and maintain RTSP connections with surveillance cameras across the campus."
multiplicity: "System connects to multiple cameras simultaneously."
primary actor: **N/A**
secondary actor: **SENSOR**::Camera::1..*
main success scenario:

1. "System authenticates Camera in the surveillance environment."
2. "System opens RTSP session with Camera using TCP or UDP transport."
3. "System receives video stream from Camera."

"System records stream metadata including camera identifier, timestamp, frame rate, GPS coordinates, and latency."

4. "System publishes frames from Camera to the message broker."

use case ends in: **SUCCESS**

extensions:

exception for 2:

2a. {**HARDWARE_EXCEPTION**::CameraOffline} #Camera is powered off or physically disconnected.#

use case ends in: **FAILURE**

exception for 3:

3a. {**NETWORK_EXCEPTION**::StreamInterrupted} #Network interruption causes stream loss from Camera.#

use case ends in: **FAILURE**

exception for 4:

4a. {**NETWORK_EXCEPTION**::MessageBrokerDown} #Message broker is unavailable. Frames from Camera cannot be published.#

use case ends in: **FAILURE**

context-awareness:

CA003: "The system records stream metadata by receiving camera ID, timestamp, frame rate, GPS coordinates, and latency from Camera."

use case: ProcessVideoFrames
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to analyse video frames from camera streams to detect security-relevant objects and events."

multiplicity: "System processes frames from multiple cameras in parallel."
primary actor: **N/A**
secondary actor: **SENSOR**::Camera
main success scenario:

"System receives frames from Camera via message broker."
"Steps 1 through 6 execute in parallel for each frame from Camera."
1. [DetectLicensePlate] "System runs license plate detection model on frame from Camera."
2. [DetectWeapon] "System runs weapon detection model on frame from Camera."
3. [DetectFireSmoke] "System runs fire and smoke detection model on frame from Camera."
4. [DetectFlood] "System runs flood detection model on frame from Camera."
5. [DetectGraffiti] "System runs graffiti detection model on frame from Camera."
6. [DetectSuspiciousBehavior] "System runs suspicious behavior detection on frame from Camera."

"System forwards all detections with metadata to the Hub of Events."
use case ends in: **SUCCESS**

extensions:

exception for (1-6):

(1-6)a. {**SOFTWARE_EXCEPTION**::DetectionModelFailure} #One or more detection models fail to process frame from Camera.#
use case ends in: **FAILURE**

context-awareness:

CA004: "The system assigns semantic meaning to detected objects by receiving frames from Camera."
CA005: "The system enriches detection events by receiving location identifier and timestamp from Camera."

use case: DetectLicensePlate
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to detect and recognize vehicle license plates from camera frames at campus entrances."
multiplicity: "System can process license plates from multiple camera feeds simultaneously."
primary actor: **N/A**
secondary actor: **SENSOR**::Camera, **SOFTWARE**::PaddleOCR

main success scenario:

1. "System receives frame from Camera positioned at campus entrance."
2. "System applies YOLOv11x plate detection model to frame from Camera." "System extracts plate region from the frame."
3. "System sends extracted plate image to PaddleOCR for text recognition."
4. "System receives recognized plate text with confidence score from PaddleOCR [DG:InterpretUnregisteredPlate]."
5. [VerifyVehicleAuthorization] "System checks recognized plate from PaddleOCR against the vehicle database."

use case ends in: **SUCCESS**

extensions:

exception for 3:

3a. {**SOFTWARE_EXCEPTION**::OCRFailure} #PaddleOCR fails to process the plate image from Camera.#

use case ends in: **FAILURE**

alternative for 4:

"Recognition confidence from PaddleOCR is below threshold of 0.85."

4a. "System logs the low-confidence detection from PaddleOCR without triggering alert [RB:InterpretUnregisteredPlate]."

use case ends in: **DEGRADED**

context-awareness:

CA006: "The system determines plate text reliability by receiving confidence score from PaddleOCR."

CA007: "The system assesses plate readability by receiving lighting conditions from Camera."

use case: VerifyVehicleAuthorization
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to verify whether a recognized vehicle is authorized to be on campus and apply pragmatic ECA rules."
multiplicity: "System can verify multiple plates concurrently."
primary actor: **N/A**
secondary actor: **SOFTWARE**::VehicleDatabase, **SOFTWARE**::SystemClock
main success scenario:

1. "System queries VehicleDatabase with recognized plate text."
2. "System receives authorization status from VehicleDatabase."
3. "System evaluates ECA rule against VehicleDatabase result and generates alert if vehicle is unauthorized [RB:InterpretUnregisteredPlate]."

"System publishes plate event with authorization status to the message broker."
use case ends in: **SUCCESS**
extensions:
exception for 1:
1a. {**SOFTWARE_EXCEPTION**::DatabaseUnavailable} #VehicleDatabase is offline or unreachable.#
use case ends in: **FAILURE**
alternative for 3:
"Vehicle is registered in VehicleDatabase."
3a. "System logs the plate event from VehicleDatabase without generating an alert [RB:InterpretUnregisteredPlate]."
use case ends in: **SUCCESS**
context-awareness:
CA008: "The system applies ECA rule by receiving authorization status and 30-day appearance frequency from VehicleDatabase."
CA009: "The system evaluates unregistered plate context by receiving current time of day and visitor hours from SystemClock."

use case: DetectWeapon
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to detect firearms or bladed weapons in camera frames and interpret the detection based on context."
multiplicity: "System processes weapon detection across multiple camera feeds in parallel."
primary actor: **N/A**
secondary actor: **SENSOR**::Camera
main success scenario:
1. "System receives frame from Camera."
2. "System applies YOLOv11x weapon detection model to frame from Camera."
3. "System identifies weapon type in frame from Camera and computes confidence score [DG:InterpretWeaponDetection]."
4. "System generates weapon detection event for Camera location with weapon type and confidence [RB:InterpretWeaponDetection]."
"System publishes weapon detection event to the message broker."
use case ends in: **SUCCESS**
extensions:
alternative for 3:
"Detection confidence from Camera frame is below alert threshold."

3a. "System logs low-confidence detection from Camera without triggering alert [RB:InterpretWeaponDetection]."
use case ends in: **DEGRADED**
context-awareness:
CA010: "The system contextualizes weapon detection by receiving person identity classification from Camera."
CA011: "The system assesses weapon detection severity by receiving location context from Camera."
CA012: "The system adjusts weapon detection urgency by receiving frame timestamp from Camera."

use case: DetectFireSmoke
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to detect fire outbreaks or smoke sources from camera frames and interpret the detection based on location context."
multiplicity: "System can detect fire and smoke from Camera across all monitored areas simultaneously."
primary actor: **N/A**
secondary actor: **SENSOR**::Camera
main success scenario:
1. "System receives frame from Camera."
2. "System applies YOLOv11x fire and smoke detection model to frame from Camera."
3. "System classifies detection in Camera frame as fire or smoke or both and computes confidence [DG:InterpretSmokeDetection]."
4. "System generates fire or smoke detection event for Camera location with classification [RB:InterpretSmokeDetection]."
"System publishes fire or smoke event to the message broker."
use case ends in: **SUCCESS**
extensions:
alternative for 3:
"Detection confidence from Camera frame is below threshold."
3a. "System logs detection from Camera for review without immediate alert [RB:InterpretSmokeDetection]."
use case ends in: **DEGRADED**
context-awareness:
CA013: "The system contextualizes smoke detection by receiving location identifier from Camera."

CA014: "The system confirms fire alert by receiving consecutive frame detections from Camera."

CA015: "The system flags potential arson by receiving frame timestamp and location from Camera."

use case: DetectFlood
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to detect flooding or water accumulation on campus from camera frames."
multiplicity: "System monitors flood-prone areas from Camera across campus."
primary actor: **N/A**
secondary actor: **SENSOR**::Camera, **SOFTWARE**::WeatherService
main success scenario:

1. "System receives frame from Camera."
2. "System applies YOLOv11x flood detection model to frame from Camera."
3. "System classifies water level in Camera frame and computes confidence."
4. "System generates flood detection event for Camera location with timestamp."

"System publishes flood event to the message broker."

use case ends in: **SUCCESS**

context-awareness:

CA016: "The system distinguishes rain from infrastructure failure by receiving current weather conditions from WeatherService."

CA017: "The system finds the flood location by receiving location identifier from Camera."

use case: DetectGraffiti
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to detect acts of graffiti or spray vandalism in progress from camera frames."
multiplicity: "System can detect graffiti from Camera across multiple monitored locations."
primary actor: **N/A**
secondary actor: **SENSOR**::Camera
main success scenario:

1. "System receives frame from Camera."
2. "System applies YOLOv11x graffiti detection model to frame from Camera."

3. "System identifies graffiti-in-progress event in Camera frame detecting person with spray can."
4. "System generates vandalism detection event for Camera location with timestamp."
"System publishes graffiti event to the message broker."
use case ends in: **SUCCESS**

context-awareness:
CA018: "The system distinguishes active graffiti from existing graffiti by receiving spray-action frames from Camera."

use case: DetectSuspiciousBehavior
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to detect suspicious human behavior near vehicles in parking areas by combining body pose analysis with vehicle data."
multiplicity: "System monitors multiple parking areas from Camera simultaneously."
primary actor: **N/A**
secondary actor: **SENSOR**::Camera, **SOFTWARE**::VehicleDatabase
main success scenario:
1. "System receives frame from Camera positioned in parking area."
2. [AnalyzeBodyPose] "System detects human keypoints in frame from Camera using YOLOv11x-Pose model."
"System tracks individual identity across frames from Camera."
"System computes spatial proximity between detected person and nearest vehicle in Camera frame."
3. "System evaluates temporal persistence of person near vehicle in Camera frame [DG:InterpretSuspiciousBehaviorNearVehicle]."
4. "System applies ECA rule to Camera data and classifies behavior based on posture, duration, and time [RB:InterpretSuspiciousBehaviorNearVehicle]."
5. "System generates suspicious behavior event for Camera location with severity level and timestamp."
"System publishes suspicious behavior event to the message broker."
use case ends in: **SUCCESS**

extensions:
alternative for 4:
"Person in Camera frame moves away from vehicle before duration threshold."
4a. "System resets tracking timer for that individual detected by Camera."
use case ends in: **SUCCESS**

alternative for 4:

"Behavior detected by Camera occurs during daytime near an authorized vehicle."

4b. "System logs the observation from Camera without raising alert [RB:InterpretSuspiciousBehaviorNearVehicle]."

use case ends in: **DEGRADED**

context-awareness:

CA019: "The system contextualizes suspicious behavior by receiving body posture classification from Camera."

CA020: "The system weights suspicious behavior by receiving frame timestamp from Camera."

CA021: "The system evaluates temporal context by receiving proximity duration data from Camera."

CA022: "The system flags suspicious behavior by receiving vehicle plate restriction status from VehicleDatabase."

CA023: "The system computes composite suspicion score by receiving spatial proximity, posture, duration, and timestamp data from Camera."

use case: AnalyzeBodyPose

scope: SurveillanceSystem

level: **SUB_FUNCTION**

intention: "System intends to detect and classify human body poses from camera frames using pose estimation."

multiplicity: "System can analyze body poses of multiple persons in a single frame from Camera."

primary actor: **N/A**

secondary actor: **SENSOR**::Camera

main success scenario:

1. "System receives frame from Camera."
2. "System applies YOLOv11x-Pose model to detect human keypoints in frame from Camera."
3. "System classifies body posture from Camera keypoints as standing, crouching, sitting, or other."
4. "System publishes pose classification data from Camera with person identifier to the message broker."

use case ends in: **SUCCESS**

extensions:

exception for 2:

2a. {**SOFTWARE_EXCEPTION**::PoseModelFailure} #YOLOv11x-Pose model fails to process frame from Camera.#

use case ends in: **FAILURE**

context-awareness:

CA024: "The system derives body posture classification by receiving human keypoint positions from Camera."

use case: DisplayAlerts
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to present detected events and alerts to operators through the centralized dashboard with deduplication."
multiplicity: "Dashboard displays alerts from all cameras simultaneously to Operator."
primary actor: **HUMAN**::Operator
secondary actor: **SOFTWARE**::HubOfEvents, **SOFTWARE**::MonitorService, **SOFTWARE**::UpdaterService, **SOFTWARE**::Dashboard
main success scenario:

1. "System receives detection event from HubOfEvents via message broker."
2. "MonitorService receives event data from System via WebSocket connection."

"MonitorService applies debounce technique: checks cache for existing event with same camera and class."

3. "MonitorService system determines the event is new and forwards it to UpdaterService via HTTP."

"UpdaterService persists event in the database."

4. "UpdaterService system publishes event to Dashboard via WebSocket."
5. "Dashboard of the system displays event to Operator with detection ID, camera location, class, priority, status, and timestamp."
6. "Dashboard of the system displays event location to Operator on geolocated map."

"Operator reviews alert and can update status, adjust priority, assign responsibility, and add notes."

use case ends in: **SUCCESS**

extensions:

alternative for 3:

"MonitorService determines the event is a duplicate within the debounce interval."

3a. "MonitorService discards the duplicate event from System."

use case ends in: **SUCCESS**

exception for 4:

4a. {**NETWORK_EXCEPTION**::DashboardUnreachable} #Dashboard client of Operator is disconnected from WebSocket.#
use case ends in: **FAILURE**
alternative for 5:
"Operator updates alert status to Closed on the Dashboard."
5a. "UpdaterService notifies MonitorService in System to remove event from cache."
use case ends in: **SUCCESS**

use case: EscalateToEmergencyServices
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "Supervisor intends to contact external emergency services when internal security response is insufficient."
multiplicity: "Supervisor can contact multiple emergency services simultaneously."
primary actor: **HUMAN**::Supervisor
secondary actor: **PHYSICAL_ENTITY**::FireDepartment, **PHYSICAL_ENTITY**::FederalPolice, **SOFTWARE**::HubOfEvents
main success scenario:
1. "Supervisor assesses incident severity on the System from dashboard data."
2. "Supervisor determines which emergency service is appropriate through System."
3. "Supervisor contacts FireDepartment or FederalPolice through the System."
4. "System logs escalation event with timestamp and service contacted by Supervisor."

use case ends in: **SUCCESS**
context-awareness:
CA025: "The system recommends appropriate emergency service by receiving incident type classification from HubOfEvents."

// Handler use cases

handler use case: HandleCameraFailure
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to manage camera failures by notifying operators and dispatching maintenance."

primary actor: **N/A**
secondary actor: **HUMAN**::Operator, **HUMAN**::ServicePerson, **SENSOR**::Camera
contexts and exceptions:
EstablishCameraConnections{**HARDWARE_EXCEPTION**::CameraOffline}
main success scenario:
1. "System detects Camera failure and logs event with camera ID and location."
2. "System sends notification to Operator on Dashboard about Camera failure."
3. "System sends maintenance request to ServicePerson with Camera location."
4. "ServicePerson inspects and repairs Camera and notifies System."
5. "System receives resolved message from ServicePerson."

"System re-establishes RTSP connection with Camera."
use case ends in: **SUCCESS**
extensions:
alternative for 4:
"ServicePerson cannot repair Camera on site."
4a. "ServicePerson replaces Camera and notifies System."
use case continues at step: 5
handler use case: HandleStreamInterruption
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to recover from network stream interruptions by re-establishing RTSP connections with Camera."
primary actor: **N/A**
secondary actor: **SENSOR**::Camera, **HUMAN**::Operator
contexts and exceptions:
EstablishCameraConnections{**NETWORK_EXCEPTION**::StreamInterrupted}
main success scenario:
1. "System detects stream interruption from Camera."
2. "System attempts to re-establish RTSP connection with Camera."
3. "System receives resumed video stream from Camera."

"System logs interruption duration and recovery timestamps."
use case ends in: **SUCCESS**
extensions:
alternative for 2:
"Reconnection attempts with Camera exceed retry limit of 5 attempts."
2a. "System marks Camera as offline and notifies Operator on Dashboard."
use case ends in: **DEGRADED**

handler use case: HandleDetectionModelFailure
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to recover when a detection model fails by restarting the model."
primary actor: **N/A**
secondary actor: **HUMAN**::Operator
contexts and exceptions:
ProcessVideoFrames{**SOFTWARE_EXCEPTION**::DetectionModelFailure}
main success scenario:
1. "System logs detection model failure and notifies Operator on Dashboard."
"System continues processing frames with remaining operational models."
2. "System attempts to restart the failed model and notifies Operator."
3. "System confirms model is operational and notifies Operator of restored capability."
use case ends in: **SUCCESS**
extensions:
alternative for 2:
"Model restart fails after 3 attempts."
2a. "System escalates failure to Operator for manual intervention."
use case ends in: **DEGRADED**

handler use case: HandleRadioCommunicationDown
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to recover from radio communication failure by using alternative channels to reach SecurityAgent."
primary actor: **N/A**
secondary actor: **HUMAN**::Operator, **HUMAN**::SecurityAgent
contexts and exceptions:
RespondToIncident{**NETWORK_EXCEPTION**::RadioCommunicationDown}
main success scenario:
1. "System detects that radio communication with SecurityAgent is unavailable."
2. "System logs radio failure event and notifies Operator on Dashboard."
3. "System attempts to reach SecurityAgent via messaging application as fallback."
4. "System receives confirmation that SecurityAgent received the notification."
use case ends in: **SUCCESS**
extensions:

alternative for 3:

"Messaging application for SecurityAgent is also unavailable."

3a. "System notifies Operator to coordinate manual dispatch of SecurityAgent."

use case ends in: **DEGRADED**

handler use case: HandleMessageServiceDown
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to recover from messaging application failure by using radio to reach SecurityAgent."
primary actor: **N/A**
secondary actor: **HUMAN**::Operator, **HUMAN**::SecurityAgent
contexts and exceptions:

RespondToIncident{**NETWORK_EXCEPTION**::MessageServiceDown}

main success scenario:

1. "System detects that messaging application for SecurityAgent is unavailable."
2. "System logs messaging failure event and notifies Operator on Dashboard."
3. "System attempts to reach SecurityAgent via radio communication as fallback."
4. "System receives confirmation that SecurityAgent received the notification."

use case ends in: **SUCCESS**

extensions:

alternative for 3:

"Radio communication with SecurityAgent is also unavailable."

3a. "System notifies Operator to coordinate manual dispatch of SecurityAgent."

use case ends in: **DEGRADED**

handler use case: HandleOCRFailure
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to recover from OCR processing failure by retrying with enhanced image from Camera."
primary actor: **N/A**
secondary actor: **SOFTWARE**::PaddleOCR, **SENSOR**::Camera, **HUMAN**::Operator
contexts and exceptions:

DetectLicensePlate{**SOFTWARE_EXCEPTION**::OCRFailure}

main success scenario:

1. "System detects that PaddleOCR failed to process plate image from Camera."
"System applies image enhancement preprocessing to the plate region."
2. "System resubmits enhanced image from Camera to PaddleOCR."
3. "System receives recognized plate text from PaddleOCR."

use case ends in: **SUCCESS**

extensions:

alternative for 2:

"Enhanced image from Camera also fails after 3 attempts with PaddleOCR."

2a. "System logs unreadable plate event from Camera and notifies Operator."

use case ends in: **DEGRADED**

handler use case: HandlePoseModelFailure
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to recover when the pose estimation model fails to process frames from Camera."
primary actor: **N/A**
secondary actor: **HUMAN**::Operator, **SENSOR**::Camera
contexts and exceptions:

AnalyzeBodyPose{**SOFTWARE_EXCEPTION**::PoseModelFailure}

main success scenario:

1. "System detects that YOLOv11x-Pose model has failed to process frame from Camera."
2. "System logs pose model failure and notifies Operator on Dashboard."
3. "System attempts to restart the pose estimation model and notifies Operator."

use case ends in: **SUCCESS**

extensions:

alternative for 3:

"Pose model restart fails after 3 attempts."

3a. "System escalates to Operator for manual intervention."

use case ends in: **DEGRADED**

handler use case: HandleDatabaseUnavailable
scope: SurveillanceSystem
level: **SUB_FUNCTION**

intention: "System intends to handle vehicle database unavailability by queuing requests and notifying Operator."
primary actor: **N/A**
secondary actor: **HUMAN**::Operator, **SOFTWARE**::VehicleDatabase
contexts and exceptions:
VerifyVehicleAuthorization{**SOFTWARE_EXCEPTION**::DatabaseUnavailable}
main success scenario:
"System detects that VehicleDatabase is unavailable."
1. "System queues pending plate verification requests for VehicleDatabase."
2. "System notifies Operator of reduced plate verification capability."
3. "System retries connection to VehicleDatabase at regular intervals."
4. "System receives confirmation that VehicleDatabase is available."
"System processes queued plate verification requests against VehicleDatabase."
use case ends in: **SUCCESS**
handler use case: HandleDashboardUnreachable
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to handle dashboard connectivity loss by buffering events until connection to Operator is restored."
primary actor: **N/A**
secondary actor: **HUMAN**::Operator, **SOFTWARE**::UpdaterService
contexts and exceptions:
DisplayAlerts{**NETWORK_EXCEPTION**::DashboardUnreachable}
main success scenario:
1. "System detects that Dashboard WebSocket connection with Operator is lost."
"UpdaterService continues persisting events to database."
2. "System retries WebSocket connection with Operator at regular intervals."
3. "System re-establishes WebSocket connection and replays buffered events to Operator."
use case ends in: **SUCCESS**

handler use case: HandleMessageBrokerDown
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to handle message broker unavailability by buffering frames from Camera locally."
primary actor: **N/A**
secondary actor: **HUMAN**::Operator, **SENSOR**::Camera

contexts and exceptions:

EstablishCameraConnections{**NETWORK_EXCEPTION**::MessageBrokerDown}

main success scenario:

1. "System detects that message broker for Camera is unavailable."
2. "System notifies Operator of reduced processing capability."

"System buffers incoming frames from Camera locally."

3. "System retries connection to message broker and notifies Operator."
4. "System re-establishes connection and flushes buffered frames from Camera."

use case ends in: **SUCCESS**

handler use case: HandleFireEmergency
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to coordinate emergency response when fire or smoke is confirmed on campus."
multiplicity: "System handles fire emergency across all affected areas."
primary actor: **N/A**
secondary actor: **HUMAN**::Operator, **HUMAN**::Supervisor, **HUMAN**::SecurityAgent, **PHYSICAL_ENTITY**::FireDepartment
contexts and exceptions:

MonitorCameraFeeds{**ENVIRONMENT_EXCEPTION**::FireDetected}

main success scenario:

mode switch: FireEmergency

"System detects fire or smoke with confidence above emergency threshold."

1. "System generates critical alert and notifies Operator on Dashboard with fire location."
2. "System sends emergency notification to Supervisor via messaging application."
3. "Supervisor assesses situation from System Dashboard data."
4. "Supervisor dispatches SecurityAgent to the affected location through System."
5. "SecurityAgent attempts to contain the situation and reports back to System."
6. "Supervisor contacts FireDepartment through System if internal response is insufficient."
7. "Supervisor sends all-clear message to System after fire is contained."

mode switch: Normal

use case ends in: **SUCCESS**

handler use case: HandleArmedThreat
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to coordinate emergency response when a weapon is detected in a threatening context."
multiplicity: "System handles armed threat across affected campus areas."
primary actor: **N/A**
secondary actor: **HUMAN**::Operator, **HUMAN**::Supervisor, **HUMAN**::SecurityAgent, **PHYSICAL_ENTITY**::FederalPolice
contexts and exceptions:
MonitorCameraFeeds{**ENVIRONMENT_EXCEPTION**::WeaponDetected}
main success scenario:
mode switch: ArmedThreatEmergency
"System detects weapon in a threatening context."
1. "System generates critical alert and notifies Operator on Dashboard with threat location and weapon type."
2. "System sends emergency notification to Supervisor via messaging application."
3. "Supervisor confirms the threat from System Dashboard data and video review."
4. "Supervisor contacts FederalPolice through System with incident details."
5. "SecurityAgent secures the perimeter and reports status to System."
6. "Supervisor sends all-clear message to System after police intervention."

mode switch: Normal
use case ends in: **SUCCESS**

handler use case: HandleFloodEmergency
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System intends to coordinate response when flooding is detected on campus."
multiplicity: "System handles flood emergency across affected campus areas."
primary actor: **N/A**
secondary actor: **HUMAN**::Operator, **HUMAN**::Supervisor, **HUMAN**::SecurityAgent
contexts and exceptions:
MonitorCameraFeeds{**ENVIRONMENT_EXCEPTION**::FloodDetected}
main success scenario:
mode switch: FloodEmergency
"System detects flooding with confidence above emergency threshold."

1. "System generates high priority alert and notifies Operator on Dashboard with flood location."
2. "System sends notification to Supervisor via messaging application."
3. "Supervisor dispatches SecurityAgent to assess flooding through System."
4. "SecurityAgent assists with evacuation and reports status to System."
5. "Supervisor coordinates cleanup and sends update to System."
6. "Supervisor sends all-clear message to System."

mode switch: Normal
use case ends in: **SUCCESS**

// CA-enabled use cases

CA-enabled use case: NighttimeSurveillanceEnhancement
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System enhances surveillance sensitivity during nighttime hours when campus activity is low and security risk is elevated."
multiplicity: "Enhancement applies across all active cameras during night shift."
context: CA002, CA012, CA020
primary actor: **HUMAN**::Operator::1..3
secondary actor: **SENSOR**::Camera::1..*
main success scenario:

1. "System detects that current time is within nighttime shift hours and notifies Operator."
2. "System lowers detection thresholds for Camera feeds on suspicious behavior and weapon detection."
3. "System elevates default alert priority for all new detections from Camera and notifies Operator."

"Operator receives enhanced alerts with elevated priority."
use case ends in: **SUCCESS**

extensions:
alternative for 1:
"Current time transitions to daytime shift."
1a. "System restores standard detection thresholds for Camera feeds and notifies Operator."
use case ends in: **SUCCESS**

CA-enabled use case: ParkingAreaIntegratedMonitoring
scope: SurveillanceSystem
level: **SUB_FUNCTION**

intention: "System provides integrated monitoring of parking areas by correlating vehicle plate recognition with person behavior detection from Camera."
multiplicity: "System monitors multiple parking areas concurrently."
context: CA019, CA020, CA021, CA022, CA008
primary actor: **HUMAN**::Operator
secondary actor: **SENSOR**::Camera::1..*, **SOFTWARE**::VehicleDatabase
main success scenario:

1. "System detects vehicles and people in parking area through Camera."
2. [DetectLicensePlate] "System identifies vehicle plates from Camera in parking area."
3. [DetectSuspiciousBehavior] "System monitors person behavior near vehicles from Camera."
   "System correlates plate authorization status from VehicleDatabase with behavior detection."
4. "System generates integrated alert to Operator if unauthorized vehicle and suspicious behavior are detected from Camera."

   use case ends in: **SUCCESS**

context-awareness:

CA026: "The system correlates vehicle authorization with suspicious behavior by receiving authorization status from VehicleDatabase."

CA-enabled use case: LocationAwareAlertPrioritization
scope: SurveillanceSystem
level: **SUB_FUNCTION**
intention: "System adjusts alert prioritization based on the type of campus location where the Camera detection occurs."
multiplicity: "Prioritization applies to all detection events across all cameras."
context: CA011, CA013, CA017
primary actor: **HUMAN**::Operator
secondary actor: **SENSOR**::Camera::1..*
main success scenario:

1. "System receives detection event with Camera location metadata."
2. "System looks up location type for Camera: sensitive area, open area, building, parking, entrance, laboratory, or kitchen."
3. "System adjusts alert priority based on Camera location type and detection type and presents to Operator."
   "Operator reviews adjusted priority on Dashboard."

   use case ends in: **SUCCESS**

context-awareness:

CA027: "The system adjusts alert priority by receiving location classification from Camera.”